1.5: Introduction to Digital Logic

1. Logic Levels

In digital electronics, **logic levels** refer to the voltage ranges that represent the two binary states: **HIGH** (1) and **LOW** (0). These levels are crucial for the proper functioning of digital circuits, as they define when a signal is interpreted as a logical '1' or a logical '0'.

For example, in Transistor-Transistor Logic (TTL) commonly used in older integrated circuits, a voltage between 2V and 5V might represent a HIGH (logic 1), while a voltage between 0V and 0.8V might represent a LOW (logic 0). The range between 0.8V and 2V is typically an undefined or transition region.

2. Logic Gates

Logic gates are the fundamental building blocks of digital circuits. They perform basic logical operations on one or more binary inputs and produce a single binary output. The output is determined by the specific logic function implemented by the gate.

2.1. AND Gate

The **AND gate** produces a HIGH output only if all its inputs are HIGH. Otherwise, the output is LOW.

Symbol:

```
A --+--\\
    |   >-- Q
B --+--//
```

Boolean Expression: $Q = A \cdot B$ or $Q = AB$

Truth Table:

Input A	Input B	Output Q
0	0	0
0	1	0
1	0	0
1	1	1

Example: Consider a security system where an alarm (Q) sounds only if both the front door sensor (A) and the window sensor (B) are triggered (HIGH). If either sensor is not triggered, the alarm remains off.

2.2. OR Gate

The **OR gate** produces a HIGH output if at least one of its inputs is HIGH. The output is LOW only if all inputs are LOW.

Symbol:

```
A --+--\\\
    |   >-- Q
B --+--///
```

Boolean Expression: $Q = A + B$

Truth Table:

Input A	Input B	Output Q
0	0	0
0	1	1
1	0	1
1	1	1

Example: Imagine a light (Q) that turns on if either switch A or switch B (or both) is flipped on. The light only stays off if both switches are off.

2.3. NOT Gate (Inverter)

The **NOT gate**, also known as an **inverter**, has a single input and a single output. It produces an output that is the inverse of its input. If the input is HIGH, the output is LOW, and vice versa.

Symbol:

```
A ---|>o--- Q
```

Boolean Expression: $Q = \bar{A}$ or $Q = A'$

Truth Table:

Input A	Output Q
0	1
1	0

Example: A warning light (Q) is on when a system (A) is off, and off when the system is on.

2.4. XOR Gate (Exclusive OR)

The **XOR gate** produces a HIGH output if its inputs are different. If the inputs are the same, the output is LOW.

Symbol:

```
A --+--\\\
    |   >-- Q
B --+--///
```

Boolean Expression: $Q = A \oplus B$

Truth Table:

Input A	Input B	Output Q
0	0	0
0	1	1
1	0	1
1	1	0

Example: A car's interior light (Q) turns on if either the driver's door (A) or the passenger's door (B) is open, but not if both are open or both are closed.

3. Boolean Algebra

Boolean algebra is a branch of algebra in which the values of the variables are the truth values `true` and `false`, usually denoted 1 and 0 respectively. It is fundamental to the design and analysis of digital circuits.

Basic Postulates and Theorems:

Postulate/Theorem	Description
$A + 0 = A$	Identity Law for OR
$A \cdot 1 = A$	Identity Law for AND
$A + 1 = 1$	Null Law for OR
$A \cdot 0 = 0$	Null Law for AND
$A + A = A$	Idempotent Law for OR
$A \cdot A = A$	Idempotent Law for AND
$A + \bar{A} = 1$	Complement Law for OR
$A \cdot \bar{A} = 0$	Complement Law for AND
$\overline{\overline{A}} = A$	Involution Law (Double Negation)
$A + B = B + A$	Commutative Law for OR
$A \cdot B = B \cdot A$	Commutative Law for AND
$A + (B + C) = (A + B) + C$	Associative Law for OR
$A \cdot (B \cdot C) = (A \cdot B) \cdot C$	Associative Law for AND
$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$	Distributive Law (AND over OR)
$A + (B \cdot C) = (A + B) \cdot (A + C)$	Distributive Law (OR over AND)
$\overline{A + B} = \bar{A} \cdot \bar{B}$	De Morgan's Theorem (OR)
$\overline{A \cdot B} = \bar{A} + \bar{B}$	De Morgan's Theorem (AND)

4. Truth Tables

A **truth table** is a mathematical table used in logic to compute the functional values of logical expressions on each of their functional arguments, that is, on each combination of values taken by their logical variables. It lists all possible input combinations and the corresponding output for a logic gate or circuit.

Example: Truth Table for a 2-input AND gate

Input A	Input B	Output Q
0	0	0
0	1	0
1	0	0
1	1	1

5. Practice Problems

Problem 1

Construct the truth table for a 3-input AND gate.

Solution:

Input A	Input B	Input C	Output Q
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Problem 2

Simplify the Boolean expression $Q = A \cdot (A + B)$.

Solution:

Using the Distributive Law: $Q = (A \cdot A) + (A \cdot B)$ Using the Idempotent Law ($A \cdot A = A$): $Q = A + (A \cdot B)$ Using the Absorption Law ($A + AB = A$): $Q = A$

Therefore, $Q = A$.

Problem 3

Given the inputs $A = 1$, $B = 0$, what is the output of an XOR gate?

Solution:

From the XOR truth table, if Input A is 1 and Input B is 0, the Output Q is 1.

References

[1] Mano, M. Morris, and Ciletti, Michael D. *Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog*. Pearson, 2018. [2] Wakerly, John F. *Digital Design: Principles and Practices*. Pearson, 2006.